A monolithic Newton–Raphson coupling of a finite-volume geometrically exact beam and a rigid body, for moored floating bodies

Paper outline

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About this outline

This is a suggested layout for writing up the monolithic beam + rigid body coupling as a paper for a numerical methods journal (for example *Int. J. Numer. Meth. Engng*, *Comput. Methods Appl. Mech. Engrg*, *J. Comput. Phys.*, *Computers & Structures*). For each section it says what should go in it, and which file in `documentation/` already holds the material.

Literature references are not listed here: each claim about prior work needs a source checked by hand before it goes in.

Alternative titles.

- Monolithic finite-volume beam–rigid body coupling with Newmark integration in displacement form
- Removing the partitioned stability limit in beam-moored rigid body simulations

Material files (all in `documentation/`):

Key	File
CP	`monolithicRigidBodyCodePath.txt`
PC	`pimpleToRigidBodySolverCallChain.txt`
ML	`monolithicMatrixLayout.txt`
TC	`twoCellMonolithicMatrix.txt`
NB	`newmarkAndMonolithicBodyRow.txt`
OC	`outerCorrectorsAndNewmark.pdf`; sources `source/outerCorrectorsAndNewmark.html`, `source/model.py`, `source/order.py`
MP	`monolithicCouplingPlan.pdf`
TR	`monolithicSolverTestProgressReport.pdf`
P5	`phase5MultiLinePlan.md`

Abstract (150–250 words)

- Problem: a floating rigid body moored by flexible lines, inside a CFD simulation. The body and lines are usually coupled by partitioned iteration.
- Difficulty: partitioned coupling becomes unstable for light bodies and stiff, short lines (the limit $\beta\,\Delta t^2 EA/(L\,m_{\mathrm{pin}}) \gtrsim 1$).

- Method: the body's Newton–Euler equations, integrated with Newmark-β in displacement form, are added as 6 extra rows and columns to the finite-volume beam's Newton system. The joint compatibility is eliminated, so beam and body are advanced together by one Newton iteration.
- Results: agrees with analytic and partitioned reference solutions; second order in time; stable where the partitioned scheme diverges; one PIMPLE outer corrector is enough, which reduces cost.
- Implementation in OpenFOAM (beamFoam / moorFV), with a single switch between partitioned and monolithic coupling.

1 Introduction

- Motivation: floating offshore structures (wave energy converters, floating wind, aquaculture) moored by lines that bend, twist and go slack. CFD of the body needs a mooring model that is both accurate and robust.
- Mooring models: quasi-static, lumped-mass (e.g. MoorDyn), finite element and finite volume beam models.
- Coupling: partitioned (body solver and line solver take turns) versus monolithic (one system). Partitioned stability problems for light bodies and stiff connections (added-mass-like instability).
- Gap: no finite-volume geometrically exact beam with the rigid body in the same Newton system (check this claim against the literature).
- Contributions:
  1. Monolithic formulation: body residuals in force units, Newmark-β in displacement form, rotations by a step rotation vector.
  2. Elimination of the joint (pin and welded) so the attachment displacement is not an unknown.
  3. Block structure of the coupled Jacobian, and its physical interpretation (action–reaction in the columns, non-symmetry).
  4. Stability analysis of the partitioned scheme on a linear model, giving the limit the monolithic scheme removes.
  5. Verification and application to moored floating bodies in waves.
- Paper layout paragraph.

2 Governing equations

2.1 Geometrically exact beam

- Total Lagrangian, Simo–Reissner type: displacement $\boldsymbol{W}$, rotation tensor $\boldsymbol{\Lambda}$ (incremental rotation $\boldsymbol{\Theta}$), internal force $\boldsymbol{Q}$ and moment $\boldsymbol{M}$.
- Balance of linear and angular momentum per unit length, including the lever-arm term $\boldsymbol{R}' \times \boldsymbol{Q}$.
- Constitutive law in section axes: $\boldsymbol{C}_Q = \mathrm{diag}(EA, GA, GA)$, $\boldsymbol{C}_M = \mathrm{diag}(GJ, EI_{yy}, EI_{zz})$. J is the torsion constant ($\pi R^4/2$ for a solid circle).
- Loads: gravity, distributed hydrodynamic load (if any), ground contact.

[material: ML section 9 (coefficient origins), TC "What each symbol is"]

2.2 Rigid body

- Newton–Euler about the centre of mass:

$$m\,\boldsymbol{a} + c\,\boldsymbol{v} = \boldsymbol{F}_{\text{ext}} + \boldsymbol{F}_{\text{beam}},$$
$$\boldsymbol{J}\boldsymbol{\alpha} + \boldsymbol{\omega} \times \boldsymbol{J}\boldsymbol{\omega} + c_r\,\boldsymbol{\omega} = \boldsymbol{M}_{\text{ext}} + \boldsymbol{r} \times \boldsymbol{F}_{\text{beam}}\ (+\,\boldsymbol{M}_{\text{beam}}).$$

- Orientation $\boldsymbol{R} = \exp([\boldsymbol{\theta}]_\times)\,\boldsymbol{R}_0$, inertia $\boldsymbol{J} = \boldsymbol{R}\,\boldsymbol{J}_{\text{body}}\,\boldsymbol{R}^T$.
- Fluid force and moment from the CFD as external loads; motion constraints (e.g. surge–heave–pitch only) as projections.

[material: ML 9.4, CP section 4]

2.3 Joint between beam end and body

- Compatibility: attachment point $\boldsymbol{u}_b = \boldsymbol{x} + (\boldsymbol{R} - \boldsymbol{I})\,\boldsymbol{r}_0$.
- Pin: force passes, no moment (zero-moment end condition).
- Welded: end section rotates with the body; force and moment pass.
- Action–reaction: the body receives $-\boldsymbol{Q}_b$ (and $-\boldsymbol{M}_b$ if welded).

[material: TC (layout, $\boldsymbol{B}_b$), ML section 6]

3 Discretisation

3.1 Finite-volume beam in space

- Cell-centred unknowns, face fluxes of force and moment.
- Newton linearisation of the face force and moment: the face coefficients $\boldsymbol{k}$, $\boldsymbol{\kappa}$, $\boldsymbol{g}$, $\boldsymbol{\gamma}$, $\boldsymbol{K}^M$, $\boldsymbol{\eta}$ (definitions and physical meaning). Boundary faces, half-cell distance Δ_p (`pDelta`).
- Pin end factor $\boldsymbol{B} = (\boldsymbol{\kappa} + \boldsymbol{\gamma})^{-1}\boldsymbol{\kappa}$: the end face rotation eliminated by the zero-moment condition.

[material: ML sections 9.1–9.3, TC "What each symbol is" and "Notation"]

3.2 Time integration

- Beam: Newmark (as implemented) with its predictor.
- Body: Newmark-β in displacement form, with predictor and acceleration

$$\tilde{\boldsymbol{x}} = \boldsymbol{x}_n + \Delta t\,\boldsymbol{v}_n + \Delta t^2\left(\tfrac{1}{2} - \beta\right)\boldsymbol{a}_n, \qquad \boldsymbol{a}_{n+1} = \frac{\boldsymbol{x}_{n+1} - \tilde{\boldsymbol{x}}}{\beta\,\Delta t^2},$$

  and the same for the step rotation vector $\boldsymbol{\theta}$. Explain why the predictor has no $\beta\,\Delta t^2\,\boldsymbol{a}_{n+1}$ term.
- Rotation Jacobian $\boldsymbol{T}(\boldsymbol{\theta})$ and its closed form.

[material: ML sections 9.4 and 10, NB sections 1 and 1a]

4 Monolithic coupling

The core of the paper.

4.1 Unknowns and residuals

- Unknowns $[\Delta \boldsymbol{W}, \Delta \boldsymbol{\Theta}]$ per cell and $[\Delta \boldsymbol{x}, \Delta \boldsymbol{\theta}]$ for the body, in 6×6 blocks.
- Body residuals in force units:

$$\boldsymbol{R}_x = \frac{m\,(\boldsymbol{x} - \tilde{\boldsymbol{x}})}{\beta\,\Delta t^2} + c\,\boldsymbol{v} - \boldsymbol{F}_{\text{ext}} + \boldsymbol{Q}_b,$$
$$\boldsymbol{R}_\theta = \boldsymbol{J}\boldsymbol{\alpha} + \boldsymbol{\omega} \times \boldsymbol{J}\boldsymbol{\omega} + c_r\,\boldsymbol{\omega} - \boldsymbol{M}_{\text{ext}} + \boldsymbol{r} \times \boldsymbol{Q}_b.$$

- Statement and short proof that a converged Newton iterate satisfies the Newmark-β update exactly (the linearisation of $\boldsymbol{Q}_b$ is the only approximation per iteration).

[material: NB sections 2–4]

4.2 Eliminating the joint

- $\Delta \boldsymbol{u}_b = \Delta \boldsymbol{x} - \boldsymbol{A}\,\Delta \boldsymbol{\theta}$, with $\boldsymbol{A} = [\boldsymbol{r}]_\times \boldsymbol{T}$.
- How the attachment-face source term $\boldsymbol{K}_b(\boldsymbol{W}_b - \boldsymbol{W}_{b,\text{prev}})$ becomes the body columns of the matrix.

[material: MP section 2, ML section 6]

4.3 Linearisation and block structure

- The coupling blocks (table), and the body diagonal including the turning-arm term $[\boldsymbol{Q_b}]_\times \boldsymbol{A}$ and the gyroscopic terms.
- Physical reading: every coupling entry is the linearised face force $\Delta \boldsymbol{Q}_b$ passed through a row operator ($\boldsymbol{I}$, $\boldsymbol{I}$, lever arm $\boldsymbol{K}^M$, $[\boldsymbol{r}]_\times$). Action–reaction appears down the columns, not across transposes.
- Why the Jacobian is not symmetric even with consistent signs.
- Pin versus welded: extra blocks for the welded joint.
- Motion constraints as projections of the body rows and columns.

[material: ML sections 1–8, TC "Where the action–reaction balance is"]

4.4 Solution of the linear system

- Assembly into one sparse matrix; sparse LU (non-symmetric).
- Sparsity: block-tridiagonal beam plus one bordering block row and column at the attachment cell. Cost relative to the beam alone.

[material: ML sections 3–4, CP sections 5–6]

4.5 Algorithm

- Pseudo-code for one time step: new time step (store state, reset $\boldsymbol{\theta}$), Newton loop (assemble beam, add body blocks, solve, update beam and body, check convergence), accept (update $\boldsymbol{v}$, $\boldsymbol{a}$, $\boldsymbol{\omega}$, $\boldsymbol{\alpha}$).

[material: CP, NB section 3]

5 The partitioned scheme and its stability limit

- The partitioned algorithm used for comparison (`FvBeamNewmark`): body Newmark step against the frozen beam force, then the beam with the end prescribed, repeated in the outer correctors.
- Linear body–spring model: amplification matrices for the partitioned loop and the monolithic update. Show the partitioned iteration diverges when $\beta\,\Delta t^2 k/m_{\mathrm{pin}} \gtrsim 1$, with $m_{\mathrm{pin}} = 1/(1/m + a^2/J)$ for an off-centre attachment.
- One outer corrector: partitioned is explicit, first order and unstable; monolithic is implicit and second order.

[material: OC (figures `growth.png`, `newton.png`), `source/model.py`, `source/order.py`, the `rigidBodyEndWhirl` sweeps]

6 Coupling with the flow solver

Keep this short: it is context, not the contribution.

- interFoam PIMPLE loop, mesh motion with `moveMeshOuterCorrectors`.
- Fluid force held fixed during the beam + body Newton loop and updated at each outer corrector (the fluid stays partitioned).
- Run-time selection of partitioned or monolithic coupling.

[material: PC, OC sections 1–2]

7 Verification

Each test: setup, what it checks, the result, a figure or table.

- Translation only: body on a beam spring, against an analytic spring–mass solution.
- Rotation: off-centre attachment and twist (`rigidBodyEndTwist`), pin and welded joints.
- Order of accuracy in time (expect second order), for both translation and rotation.
- Newton convergence: quadratic convergence of the residual norm (show the exact Jacobian pays off).
- Monolithic vs converged partitioned on cases where both run: they should agree to the iteration tolerance.
- Robustness: `rigidBodyEndWhirl` sweeps of moment and EI; the partitioned limit seen numerically, matching the stability section.

[material: TR, `tutorials/rigidBodyEnd*`, `rigidBodyAndBeam_loop` and `_beamFoamCoupled`, the regression check of 2026-10-20 (agreement to $\sim 2 \times 10^{-6}$ with identical Newton counts after the legacy removal)]

8 Application: moored floating bodies in waves

- `floatRigidBody`: floating box with a mooring line in waves; motion and line force, monolithic vs partitioned vs MoorDyn.
- Slack line (`floatRigidBody_slackLine`): snap loads and compression, where line bending and ground contact matter.

- Two lines (`floatRigidBody_twoLine`), if Phase 5 is complete.
- Sweeps of line stiffness and wave period (`floatRigidBody_sweeps`): where the partitioned scheme fails, and cost.
- Cost: Newton iterations per step (`beamSolverIterations` function object), outer correctors needed, wall time split fluid / beam.

[material: `tutorials/floatRigidBody*`, `plot_floatRigidBody_sweeps.py`, `plot_monolithic_vs_partitioned_*.py`]

9 Discussion

- When monolithic coupling matters (the stability number) and when partitioned is good enough.
- Cost: the extra rows are cheap; the saving comes from fewer outer correctors and fewer failed runs.
- Limitations:
  - one line per body today (Phase 5 plan for several lines);
  - the fluid stays partitioned (no fluid added mass in the monolithic system);
  - non-symmetric Jacobian, so a direct sparse LU; scaling to long lines and many lines;
  - geometric terms that are only first order in $\boldsymbol{\theta}$ ($\boldsymbol{T} = \boldsymbol{I}$ limit).
- Extensions: several lines on one body, several bodies, implicit fluid added mass, iterative solvers with block preconditioners.

[material: P5, OC]

10 Conclusions

- Restate the contributions and the main quantitative results (the stability number, order of accuracy, cost).

A The full two-cell matrix in physical terms

With the meaning of every entry.
[material: TC]

B Derivation of the face coefficients

$\boldsymbol{k}$, $\boldsymbol{\kappa}$, $\boldsymbol{g}$, $\boldsymbol{\gamma}$, $\boldsymbol{K}^M$, $\boldsymbol{\eta}$, and the pin factor $\boldsymbol{B}$.
[material: ML section 9, TC "Notation"]

C Rotation

Exponential map, rotation Jacobian $\boldsymbol{T}(\boldsymbol{\theta})$, and the rotational inertia Jacobian.
[material: ML 9.4]

D Linear stability model

Of the partitioned and monolithic schemes.
[material: OC, `source/model.py`]

E Implementation notes and code availability

Repositories, branch, and the tutorial cases that reproduce each figure.
[material: CP, PC]

Figures and tables to prepare

Item	Content
Fig. 1	Sketch: body, joint, beam cells and faces, with $\boldsymbol{r}$, $\boldsymbol{Q}_b$, $\boldsymbol{x}$, $\boldsymbol{\theta}$.
Fig. 2	Sparsity pattern of the coupled Jacobian (beam blocks + bordered body block).
Fig. 3	Linear model: spectral radius vs $\beta\,\Delta t^2 k/m_{\text{pin}}$, partitioned and monolithic (`growth.png`).
Fig. 4	Order of accuracy in time.
Fig. 5	Newton residual vs iteration (quadratic convergence).
Fig. 6	Verification time histories against analytic / reference.
Fig. 7	Robustness map: where partitioned diverges, monolithic runs.
Fig. 8	`floatRigidBody` motion and line force: monolithic, partitioned, MoorDyn.
Fig. 9	Cost: Newton iterations and wall time per step, sweeps.
Tab. 1	Notation.
Tab. 2	Coupling blocks and their physical meaning.
Tab. 3	Case parameters.

Gaps to fill before writing

- A clean order-of-accuracy study in the full code (not only the linear model), for translation and rotation.
- A Newton convergence plot from the code showing quadratic convergence, with and without the welded terms.
- Wall-time measurements on the same machine for monolithic vs partitioned with the outer corrector count each needs.
- A literature review to support the "gap" claim in the introduction and to place the method against monolithic FSI and mooring coupling work.